

Kausar Patherya

kpatherya3@gatech.edu | kausarpatherya.com | scholar.google.com | linkedin.com/in/kpatherya | github.com/kpatherya

EDUCATION

Georgia Institute of Technology <i>Ph.D. in Computer Science, GPA: 3.96/4.00</i>	Atlanta, GA <i>Jan. 2026 – Present</i>
Georgia Institute of Technology <i>M.S. in Computer Science, GPA: 3.95/4.00</i>	Atlanta, GA <i>Aug. 2024 – May 2026</i>
University of Illinois at Urbana-Champaign <i>B.S. in Industrial Engineering, GPA: 3.79/4.00</i>	Champaign, IL <i>Aug. 2018 – May 2022</i>

EXPERIENCE

Doctoral Researcher <i>LLAMAS Lab, Georgia Tech (Advisor: Prof. Francisco Romero)</i>	Oct. 2025 – Present <i>Atlanta, GA</i>
- Built Flash-Fusion, an edge-cloud system that compiles natural-language questions over IoT sensor streams into schema-aware typed plans, gates them on live-schema validation, and runs deterministic pandas operators that emit per-query routing, plan, cache, and execution evidence. - Cut query latency by 95% and token usage and cost by 98% against a raw-telemetry LLM baseline by pairing edge statistical summarization with cloud-side query planning; deployed and evaluated on a university bus fleet.	
Graduate Research Assistant <i>WiSciTech Lab, Georgia Tech (Advisor: Prof. Ashutosh Dhekne)</i>	Aug. 2024 – Sep. 2025 <i>Atlanta, GA</i>
- Reduced cellular payload for fleet monitoring by 73.5% by architecting a tiered IoT pipeline, writing Zephyr RTOS firmware for nRF9160 boards that aggregates inertial data on-device before LTE transmission (github.com/kpatherya/bus-rtos). - Built Sting-Sense, an interactive analytics layer over that fleet telemetry; applied K-means clustering to identify five driving-behavior profiles and geographic safety hotspots (sting-sense.vercel.app).	
Host & Producer <i>Georgia Tech AI Podcast</i>	Dec. 2024 – Feb. 2026 <i>Atlanta, GA</i>
Produced 14 episodes interviewing Georgia Tech AI faculty on robotics, continual learning, and trust in autonomy; handled recording, editing, and publication to YouTube and Spotify (georgiatech.ai).	
Technology Consultant <i>Ernst & Young</i>	Jul. 2022 – May 2024 <i>Chicago, IL</i>
- Developed an RPA fraud-detection bot for a \$16B fintech client, saving 135 annual productivity hours and supporting a projected five-year ROI of \$600K. - Led cloud-delivery work spanning Azure region selection, Microsoft 365 tenant discovery, and AWS serverless application development with Terraform and GitHub Actions.	

PUBLICATIONS

Flash-Fusion: Enabling Expressive, Low-Latency Queries on IoT Sensor Streams with LLMs. Kausar Patherya , Ashutosh Dhekne, Francisco Romero. Preprint, Nov. 2025; under review. arxiv.org/abs/2511.11885 github.com/kpatherya/flash-fusion flash-fusion.vercel.app

SELECTED PROJECTS

POLARIS <i>OWL-ViT, FastVLM, PyTorch, OpenCV</i>	github.com/kpatherya/polaris
Designed a staged vision-language pipeline that matches landmarks across seasonal change, chaining histogram keyframe selection, open-vocabulary detection, semantic enrichment, depth consistency, and RANSAC verification to keep 50k-image datasets tractable.	
Aerial-Ground Robot Platform <i>ROS 2, VINS-Mono, RealSense, OctoMap</i>	github.com/kpatherya/quad-ugv-mapping
Integrated a reproducible ROS 2 visual-inertial mapping toolbox for aerial and ground robots, fusing VINS-Mono and RealSense depth through TF-aligned point clouds into OctoMap and validating trajectories on EuRoC sequences.	
AutoCaLC <i>CausalWorld, Stable-Baselines3, reinforcement learning</i>	github.com/kpatherya/autocalc
Developed a meta-RL curriculum framework in which a tabular Q-learning teacher uses UCB exploration over validation learning-progress rewards to sequence environment interventions for TriFinger manipulation tasks.	

TECHNICAL SKILLS

Programming: Python, SQL, C++, TypeScript
Frameworks & Tools: PyTorch, pandas, Pydantic, scikit-learn, ROS 2, Zephyr RTOS, Docker, AWS, Terraform
Research Areas: LLM agents, AI systems, data systems, sensor analytics, edge AI, vision-language models, reinforcement learning